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To cite this article: Kun Hu and Yuanbin Mo 2024 *Meas. Sci. Technol.* **35** 046203

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A novel unmanned aerial vehicle path planning approach: sand cat optimization algorithm incorporating learned behaviour

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Received 8 October 2023, revised 23 November 2023

Accepted for publication 28 December 2023

Published 24 January 2024



CrossMark

Abstract

Unmanned aerial vehicle (UAV) path planning plays an important role in UAV flight, and an effective algorithm is needed to realize UAV path planning. The sand cat algorithm is characterized by simple parameter setting and easy implementation. However, the convergence speed is slow, easy to fall into the local optimum. In order to solve these problems, a novel sand cat algorithm incorporating learning behaviors (LSCSO) is proposed. LSCSO is inspired by the life habits and learning ability of sand cats and incorporates a new position update strategy into the basic Sand Cat Optimization Algorithm, which maintains the diversity of the population and improves the convergence ability during the optimization process. Finally, LSCSO is applied to the challenging UAV 3D path planning with cubic B-spline interpolation to generate a smooth path, and the proposed algorithm is compared with a variety of other competing algorithms. The experimental results show that LSCSO has excellent optimization-seeking ability and plans a safe and feasible path with minimal cost consideration among all the compared algorithms.

Keywords: path planning, unmanned aerial vehicle, sand cat optimization algorithm incorporating learned behaviour, sand cat optimization algorithm

1. Introduction

In recent years unmanned aerial vehicle (UAV) has been widely used in agriculture [1], environmental monitoring [2], construction [3], logistics and transportation [4], disaster response [5], scientific exploration [6], entertainment and media [7], and many other fields. Particularly noteworthy is that UAV path planning, as an important research direction, has a direct impact on the practical effects and performance optimization of UAVs in several fields.

UAV path planning is a typical complex multi-objective optimization problem. As the dimension of the planning problem increases, the difficulty of solving it grows exponentially. In order to solve the trajectory planning problems under different conditions, scholars have continuously studied and improved the trajectory planning algorithms.

There are two types of typical methods, one is the traditional optimization methods, such as Dijkstra's algorithm [8], A* algorithm [9], artificial potential field method [10], etc; the other is the swarm intelligence optimization algorithms, such as the particle swarm algorithm [11], genetic algorithm [12], ant colony optimization algorithm [13], grey wolf optimization algorithm [14], Search and rescue optimization algorithm

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[15], artificial bee colony algorithm [16], and differential evolution algorithms [17], etc which have all been used in solving the trajectory planning problems. However, since the no free lunch [18] approach claims that there is no particular algorithm that provides the best solution for every optimization problem, it is natural to develop new metaheuristic algorithms that can be used in a variety of problems.

Sand cat optimization algorithm (SCSO) is a newly proposed algorithm in 2023 [19], which has attracted much attention due to its simple parameter setting and easy implementation. However, it still has some defects [20, 21]. The algorithm is mainly divided into two phases: the exploration phase and the exploitation phase. The former is used for global search and the latter is used for local search. Compared with other meta-heuristic algorithms, SCSO proposes a new strategy to update individual positions, i.e. the current individual may not move according to the global optimal position, but make position changes by communicating with random candidates. However, this strategy leads to slower convergence of the algorithm and poor accuracy of the optimization search. Therefore, this paper proposes a new learning behavior-based sand cat algorithm (LSCSO), which accelerates the convergence speed of the sand cat by introducing elite candidates and proposes a new position update model by mimicking the learning behavior of the sand cat. The experimental data show that LSCSO improves the convergence speed, overcomes the disadvantage of poor accuracy in finding the best, and has a positive impact on path planning. In addition, the use of 3 times B-spline [22] interpolation to smooth the path can better meet the performance constraints of UAVs.

The main contributions of this paper are summarized below:

- It solves the problem that the original algorithm is slow to converge and easy to falls into the local optimum.
- It solves the UAV path planning Problem with minimal cost consideration and performs better than other well-known algorithms.
- A new update agent location model inspired by sand cat learning behavior is proposed.

This paper's remaining sections are structured as follows: section 2 introduces the UAV flight environment model and the cost constraints; section 3 discusses the basic sand cat algorithm; section 4 presents the enhanced sand cat algorithm based on learned behaviors; section 5 explains experimentation and analysis of the UAV path planning problem; The conclusion and future works take place in the last section.

2. Mathematical modeling of UAV path planning

2.1. Terrain environment model

UAVs must acquire terrain information quickly and accurately in order to carry out safety surveys, field rescues, and other tasks. Good terrain modeling can significantly improve the effectiveness of path planning. In this paper, we refer to the

environment model in the literature [23] for modeling, and its baseline terrain model is:

$$Z(x, y) = \sum_{i=1}^n h_i \exp \left[- \left(\frac{x - x_i}{x_{si}} \right)^2 - \left(\frac{y - y_i}{y_{si}} \right)^2 \right]. \quad (1)$$

Where (x_i, y_i) is the center coordinate of the i th peak; h_i is a terrain parameter, which controls the height of the peak; x_{si} and y_{si} is the attenuation of the i th peak along the x -axis and y -axis directions, respectively, which is used to control the slope of the peak; n denotes the total number of peaks, and its environment model diagram is shown in figure 1:

2.2. Drone flight path representation

A cubic B-spline smoothing curve [22] is used to associate individual spatial coordinate locations adjacent to one another in the UAV's flight path, which is represented by ordered point coordinates [24]. Assume that the set of node sequences is $\{S, W_1, W_2, \dots, W_{n-1}, G\}$, this set of node sequences consists of $N+1$ nodes; S and G represent the starting point and the ending point of the UAV respectively; $W_1 \dots W_{n-1}$ are the individual nodes in the flight process; Set the three-dimensional representation of the takeoff point and the termination point is $S = (x_0, y_0, z_0)$, $G = (x_n, y_n, z_n)$, and describe each intermediate node by $W_i = (x_i, y_i, z_i)$ ($i \dots n-1$) to describe each intermediate node.

2.3. Restrictive condition

Path planning for UAVs has many constraints that limit the maneuvering actions of the UAV, making it very difficult to find the optimal path. The purpose of constraints is to ensure that flyable paths are planned, and in this paper, two constraints, terrain, and environmental constraints are proposed for single UAV path planning.

In order to avoid collision when accomplishing the information acquisition task, the UAV flight altitude should always be higher than the terrain altitude, so a suitable minimum altitude constraint needs to be selected. The UAV will be required to always satisfy the:

$$Z_i > Z_{\min}(x_i, y_i), i = 1, 2, \dots, n \quad (2)$$

where $Z_{\min}(x_i, y_i)$ is the terrain function that returns the terrain height at $S = (x_i, y_i)$ and Z_i is the UAV flight height.

In the process of performing the information acquisition task, to plan a better path and at the same time reduce the cost, it is stipulated that the UAV can only work in a specified area and the environmental constraints are modeled as:

$$\begin{cases} 1 \leq x_i \leq x_{\max} \\ 1 \leq y_i \leq y_{\max} \\ 20 \leq z_i \leq z_{\max} \\ i = 1, 2, \dots, n. \end{cases} \quad (3)$$

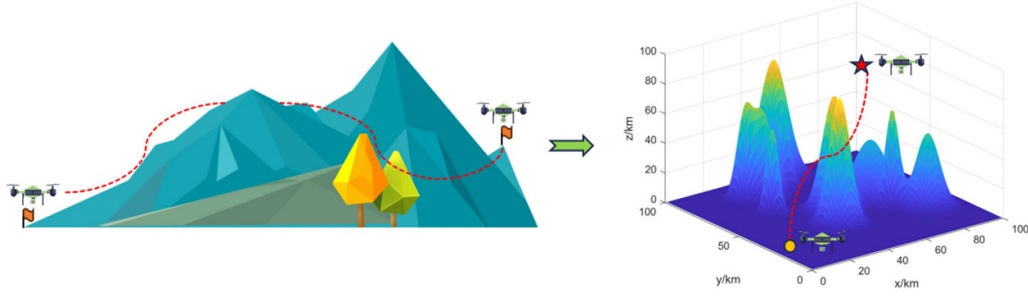


Figure 1. Terrain environment model.

2.4. Cost function

The cost function is used to evaluate the degree of merit of the generated paths, in order to better assess the quality of the paths, this paper comprehensively considers the three aspects of the path's length cost, terrain cost, and the size of the boundary cost to construct the fitness function.

The path cost P_c mainly considers the length of the UAV flying from the starting point to the endpoint and is proportional to the distance L . If the total trajectory consists of n waypoints, then the total path cost can be expressed as follows:

$$P_c = \sum_{i=1}^{n-1} L_i. \quad (4)$$

The terrain cost T_c mainly considers the threat of mountain peaks during the UAV's information acquisition task, and this cost constraint enables the UAV to avoid obstacles during the work process with the following expression:

$$\begin{cases} T_{c0} = 0 \\ T_c = \sum_{i=1}^n T_{ci} \\ T_{ci} = \begin{cases} M, & z_i < z_{\min}(x_i, y_i) \\ 0, & \text{others} \end{cases} \end{cases} \quad (5)$$

where the initial terrain cost $T_{c0} = 0$, T_c is the cumulative cost of each segment, when the UAV's altitude z_i is lower than the terrain altitude $z_{\min}$, it means that a collision occurs, at this time, the terrain cost of the segment T_{ci} is assigned as M ($M > 0$), otherwise it is 0.

The boundary cost E_c is mainly considered to ensure that the UAV works in the specified spatial region during the information acquisition task, and the expression is as follows:

$$\begin{cases} E_{c0} = 0 \\ E_c = \sum_{i=1}^n E_{ci} \\ E_{ci} = \begin{cases} M, & x_i \notin [0, x_{\max}] \parallel y_i \notin [0, y_{\max}] \parallel z_i \notin [0, z_{\max}] \\ 0, & \text{others} \end{cases} \end{cases} \quad (6)$$

where the initial boundary cost $E_{c0} = 0$, E_c is the cumulative boundary cost of each segment, and when the UAV exceeds the flight boundary, the boundary cost of the segment E_{ci} is assigned to M , otherwise, it is 0.

In summary, for flyable paths, the UAV's path length cost, obstacles, and boundary constraints are considered together,

and the following expression for the UAV's integrated cost function can be generalized:

$$\min(W) = \min(P_c + T_c + V_c) \quad (7)$$

where P_c denotes path length cost; T_c denotes terrain cost; E_c and denotes boundary cost; W is the total cost.

3. The basic SCSO

The Sand Cat is a type of cat that lives in the harsh environments of sandy and stony deserts. It can detect low frequencies below 2k Hz, and thanks to this characteristic, the sand cat has an amazing ability to search for prey. The SCSO algorithm consists of two phases: a search phase and a predation phase. In addition, a mechanism to balance exploration and exploitation is proposed.

3.1. Search phase

The prey search mechanism of the sand cat relies on low-frequency noise emission. Sand cats can perceive low frequencies below 2k Hz, and in the mathematical model, this value will be linearly reduced from 2 to 0 as the iterative process proceeds to gradually approach the prey without losing or skipping it. The auditory properties of the sand cat are simulated with a value of 2 and its mathematical model is:

$$r_G = S_M - \left(\frac{S_M \times t}{T} \right). \quad (8)$$

Where t is the current number of iterations; T is the maximum number of iterations. In the search step, the position update of each current search agent is based on a random position and the search agent is able to explore a new space in the search space. To avoid falling into a local optimum, each sand cat has a different sensitivity range and its range is:

$$r = r_G \times \text{rand}(0, 1). \quad (9)$$

Each sand cat updates its position based on the optimal solution position Pos_{bc} , its own current position Pos_c , and its sensitivity range r . Thus the sand cat can find other best prey positions as in equation (10), this equation allows the

algorithm to find a new local optimum, which is located between the current position and the prey position.

$$\text{Pos}(t+1) = r \cdot (\text{Pos}_{bc}(t) - \text{rand}(0,1) \cdot \text{Pos}_c(t)). \quad (10)$$

3.2. Attack phase

To simulate the process of a sand cat attacking its prey, the distance Pos_{md} between the sand cat and its prey is calculated by equation (11). Meanwhile, assuming that the sensitivity range of the sand cat is a circle, the direction of movement utilizes a roulette wheel selection algorithm to select a random angle θ for each sand cat. Since the selected random angle is between 0 and 360 degrees, each member of the group is able to move along a different circumferential direction in the search space, and after that, to attack the prey according to equation (12). With this method, sand cats can approach the hunting position faster.

$$\text{Pos}_{\text{md}} = |\text{rand}(0,1) \cdot \text{Pos}_b(t) - \text{Pos}_c(t)| \quad (11)$$

$$\text{Pos}(t+1) = \text{Pos}(t) - r \cdot \text{Pos}_{\text{md}} \cdot \cos(\theta). \quad (12)$$

3.3. Balancing mechanism

The main parameter controlling the transition between the search and predation phases is R , which allows the SCSO to switch between the two phases. When $|R| \leq 1$, then the position is updated according to equation (12), otherwise, the task of the sand cat is to search for a new possible solution globally and update the position according to equation (10). The mathematical model of the smoothing mechanism is as follows:

$$R = 2 \times r_G \times \text{rand}(0,1) - r_G \quad (13)$$

$$X(t+1) = \begin{cases} \text{Pos}(t) - r \cdot \text{Pos}_{\text{md}} \cdot \cos(\theta), & |R| \leq 1 \\ r \cdot (\text{Pos}_{bc}(t) - \text{rand}(0,1) \cdot \text{Pos}_c(t)), & |R| > 1. \end{cases} \quad (14)$$

4. Sand cat optimization algorithm for incorporating learned behavior

4.1. Refractive learning

It is well known that the sand cat is able to search for prey because it is able to detect low frequencies below 2k HZ and locate the prey by capturing the sound it makes. However, in different environments and air qualities, sound waves emitted by prey travel in different directions in the medium. On sandy days, when the sand content in the air is high, low-frequency sound waves propagate in such environments with refraction. The principle is shown in figure 2.

Take two-dimensional space as an example, $[lb, ub]$ is the search range of the solution, and the y -axis represents the propagation medium. Suppose there is a prey S , the height is h , the projection on the x -axis is x ; the low-frequency sound wave emitted by this prey undergoes the refraction phenomenon in the propagation medium, and after passing through the medium y , it propagates to the other side of the position S^* ,

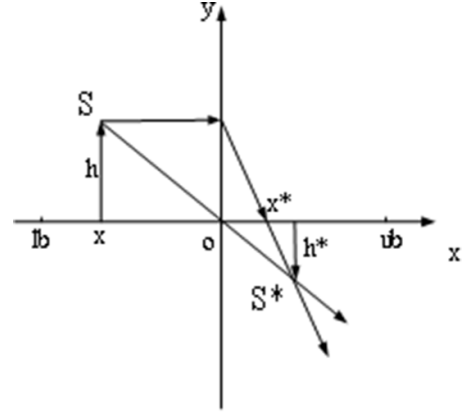


Figure 2. Refraction diagram.

the height is h^* , and the projection on the x -axis is x^* . The principle of refraction of sound waves can be obtained:

$$\frac{(lb+ub)/2 - x}{x^* - (lb+ub)} = \frac{h}{h^*}. \quad (15)$$

Let $k = h/h^*$, then equation (15) can be rewritten as:

$$x^* = \frac{lb+ub}{2} + \frac{lb+ub}{2k} - \frac{x}{k}. \quad (16)$$

Equation (16) is the solution formula for the inverse solution of the acoustic refraction learning strategy. When $k = 1$, equation (16) can be simplified as:

$$x^* = lb + ub - x. \quad (17)$$

This equation is the solution formula for inverse learning [25]. From the above, it can be seen that inverse learning is special refractive learning. By adjusting the size of k , dynamically changing solutions can be obtained in refractive learning, which further improves the algorithm's optimization-seeking ability and expands the search range of the sand cat. In this paper, the value of k is defined as the refraction coefficient and the formula for its value is as follows:

$$k = \left(1 + \left(\frac{3 \times \text{iter}}{\text{iter_max}} \right)^{\frac{1}{2}} \right)^8. \quad (18)$$

4.2. Sand cats learning behavioral strategies

Sand cats are desert survival experts, mainly found in central Asia. Due to the relatively low species diversity in their environment, sand cats have much fewer food options than other felines. It can endure in such a hostile environment thanks to its excellent learning capacity as well as its predatory abilities. In this research, we suggest a sand cat learning behavior method that may efficiently improve the local exploitation capability of the algorithm, and its position update formula is shown in equation (19):

$$X_i(t+1) = X_i(t) + L * \text{Step}. \quad (19)$$

Where t denotes the current iteration number, $X_i(t)$ and $X_{i+1}(t)$ denote the before and after position of the sand cat individual respectively. $Step$ denotes the growth value. L is the learning factor, which represents the conversion rate of the knowledge acquired by the individual sand cat during the learning process. The equation is as follows

$$L = 0.2 + 0.8 * \text{rand}. \quad (20)$$

Two operators, ‘imitation learning’ and ‘perceptual learning’, are included in the above learning behavior strategy for improving the cat’s optimization and global search ability.

4.2.1. Imitation learning factor. Sand cats like to prey on white mice and trap them by camouflaging them in their environment. The sand cat’s ability to hunt for food is learned through later observation and imitation of its counterparts, which reflects the sand cat’s strong ability to learn by imitation, an ability that is important in adapting to its environment. Therefore, the ‘imitation learning’ factor was proposed. The new state of the sand cat after learning depends on the merit of the location of the learning object. The formula is as follows:

$$\text{Step} = \begin{cases} X_b - X_p, f(X_b) \leq f(X_p) \\ X_p - X_b, f(X_b) > f(X_p) \end{cases} \text{rand} \leq 0.5 \quad (21)$$

where X_b and X_p denote the location of the individual sand cat and the learning object, respectively.

4.2.2. Perceptual learning factor. ‘Perceptual learning’ refers to the ability of animals to learn to adapt and solve problems by observing and perceiving information about their surroundings, and is one of the most important means of ensuring the continuity of populations. For example, seagulls seem to be able to sense changes in air pressure and migrate quickly to safety before a storm hits.

Sand cats are no exception. They can accurately sense the low frequencies generated before an incoming sandstorm, and once the danger is detected, sand cat colonies spread this danger signal to each other, guiding the population to migrate to safety. Perceptual learning ability is necessary for sand cat populations to survive in harsh environments such as deserts. Inspired by this, we propose the ‘perceptual learning’ factor:

$$\text{Step} = X_{c1} - X_{c2}, \text{rand} > 0.5 \quad (22)$$

where X_{c1} and X_{c2} are any two sand cats in the population that perform learning and share perceptual information.

4.3. Elite candidate learning

In the early iterations of the original sand cat algorithm, the population position is updated by selecting the optimal candidate sand cat, but essentially using a random number generator to select a candidate sand cat from all populations. This has a partial probability of selecting a poorly adapted population such that the algorithm struggles to perform in the early iterations. To address this problem, this paper proposes an elite

candidate learning method, i.e. constructing an elite candidate pool, selecting the top 10% of adapted populations in each iteration as elite candidates, and updating the positions of the populations according to equation (10). The elite pool is modeled as follows:

$$\text{Pool}(X_1, X_2, \dots, X_{0.1N}). \quad (23)$$

4.4. Algorithm execution steps

The steps of LSCSO and the associated pseudo-code are as follows:

Algorithm 1. The LSCSO optimization algorithm.

```

1 Random initialization of populations in [lb,ub];
2 Generate refractory populations according to equation (16),
  retaining well-adapted;
3 while  $t < T$  do
4   for  $i = 1 : N$  do
5     if  $|R| \leq 1$  then
6       Update the population position according to
        equation (12);
7     else
8       Elite candidate pools are created according to
        equation (23) and population positions are updated
        according to equation (10).
9     end if
10  end for
11  if  $\text{rand} \leq 0.5$  then
12    Update the position according to the ‘Imitation learning’
    operator in (21);
13  else
14    Update the position according to the ‘perception learning’
    operator in (22);
15  end if
16  Calculate the updated fitness and keep it if it is better than the
    original fitness, otherwise delete it.
17   $t = t + 1$ 
18 end while

```

4.5. Time complexity analysis

Time complexity is one of the important indicators of the efficiency of an algorithm. From the literature [19], the time complexity of SCSO is $O(n \times m)$, where n represents the number of populations and m represents the problem size. Assuming that $n = m$, the time complexity of the SCSO algorithm is $O(n^2)$. For LSCSO, the time complexity of population initialization is $O(n \times m)$. The time complexity of refractive learning and selecting the better population is $O(1)$. In addition, the time complexity in the position updating process of the agent population is $O(n \times m)$. The time complexity of updating the position of the population through ‘imitation learning’ and ‘perception learning’ is $O(1)$. The time complexity of creating the elite candidate pool and selecting the candidate population is $O(1)$. According to the algorithmic steps in the literature [19], the time complexity of the new and improved LSCSO algorithm is computed in the form of $O(n \times m + 1 + n \times m + 1 + 1)$. When $n = m$, the time complexity is $O(2n^2) = O(n^2)$.

It can be concluded that the time complexity of the improved algorithm is of the same order of magnitude as that of the original algorithm and that the time complexity of this algorithm is closely related to the population size and the problem size.

5. Experimental simulation and result analysis

5.1. Experimental environment and parameter settings

Considering that the index of the algorithm running time is largely related to the performance of the computer's processor. So the operating system of the test environment of this simulation is unified as Windows, CPU is 12th Core i5-12400F, memory 16G, main frequency 2.5 GHz, and simulation software is Matlab2022a.

In this paper, the classical particle swarm optimization algorithm (PSO) [26], genetic algorithm (GA) [12], ant colony optimization algorithm (ACO) [27], artificial bee colony optimization algorithm (ABC) [16], and A* algorithm are selected to conduct comparative experiments with the algorithm before and after the improvement, in order to validate the reasonableness of the improvement strategy.

Considering the influence of the algorithm parameters on the experimental effect, in order to avoid the sensitivity of the algorithm parameters, the parameters in the original text are used directly for the simulation experiments to ensure the rigor of the experiments to the greatest extent. The specific parameter settings of each algorithm are shown in table 1. Based on the objectivity and authenticity of the simulation test, all the algorithms are set to 30 populations, the maximum number of iterations is 50, the execution space is 100*100*100, and each experiment is run independently 10 times to record the optimal value, the average value and the standard deviation of the results.

5.2. Comparison experiments with other algorithms

5.2.1. Low-dimensional experiments and analysis.

Experiment 1 first used a 4-peak terrain to verify the optimization finding ability of the improved algorithm. The starting point was set as (0, 0, 20) and the endpoint as (100, 100, 80). Let the straight line distance between the two points be the theoretical optimum, then the ideal optimum of experiment 1 is 153.62. In order to check the experimental error and better verify the advantages of LSCSO, the error rate ω is defined in this paper:

$$\omega = \frac{l_a - l_i}{l_i}. \quad (24)$$

where l_a is the optimal value for the actual flight and l_i is the theoretical optimal value. For low-dimensional experiments, the experimental errors for each algorithm are shown in table 8. The experimental data show that LSCSO has the lowest error rate, reflecting the superiority of the improved algorithm.

Table 1. Parameters of each algorithm.

Algorithm	Parameter	Value
PSO	Inertia weight (ω)	0.9
	Social weight (c_1)	2
	Cognitive weight (c_2)	2
GA	Chromosome length	3
	Selection probability	0.5
	Crossover probability	0.8
	Mutation probability	0.2
ABC	Number of bees	30
	Honey bees	15
	Observe bees	15
	Number of nectar sources	15
ACO	Pheromone importance factor (α)	10
	Heuristic factor (β)	1
	Pheromone volatilization factor (γ)	0.1
	Constant (ϑ)	1
SCSO	Sensitivity (S_M)	2
LSCSO	Sensitivity (S_M)	2

Table 2. Experimental data.

Algorithm	Statistical data			Wilcoxon test		
	Avg	Opt	Std	P-value	Sign	Time
PSO	158.89	158.54	1.28	0.0440	+	9.46
GA	187.15	172.35	11.63	0.0002	+	11.29
A*	—	159.59	—	—	—	16.89
ABC	160.11	154.39	5.77	0.0427	+	10.03
ACO	203.23	197.65	10.33	0.0002	+	38.88
SCSO	166.03	155.28	8.39	0.0312	+	11.85
LSCSO	157.38	153.87	2.62	—	—	9.34

In order to verify the significant difference between the LSCSO algorithm and other competing algorithms, we analyzed the significance of LSCSO using the nonparametric Wilcoxon rank sum test [28]. The significance level p is set to 0.05. Table 2 succinctly lists the average fitness, average running time, and rank-sum test of the various algorithms. It is worth noting that the symbol '+' denotes a significant difference, where the LSCSO algorithm is statistically significantly better than its contemporaries, while '-' denotes its antithesis. From the results of the table, it can be seen that the LSCSO algorithm shows significant performance improvement and shorter average CPU time as compared to other algorithms.

Subfigures (a)–(c) in figure 3 shows the 3D view, front view, and top view, respectively, of all the algorithms planning paths in Experiment 1. Subfigures (d)–(f) show the average fitness of several heuristic algorithms, the iteration curves, and the fitness box-and-line plots for ten experiments, respectively. As can be seen from the figure, all algorithms plan feasible flight paths. It is noteworthy that LSCSO plans a

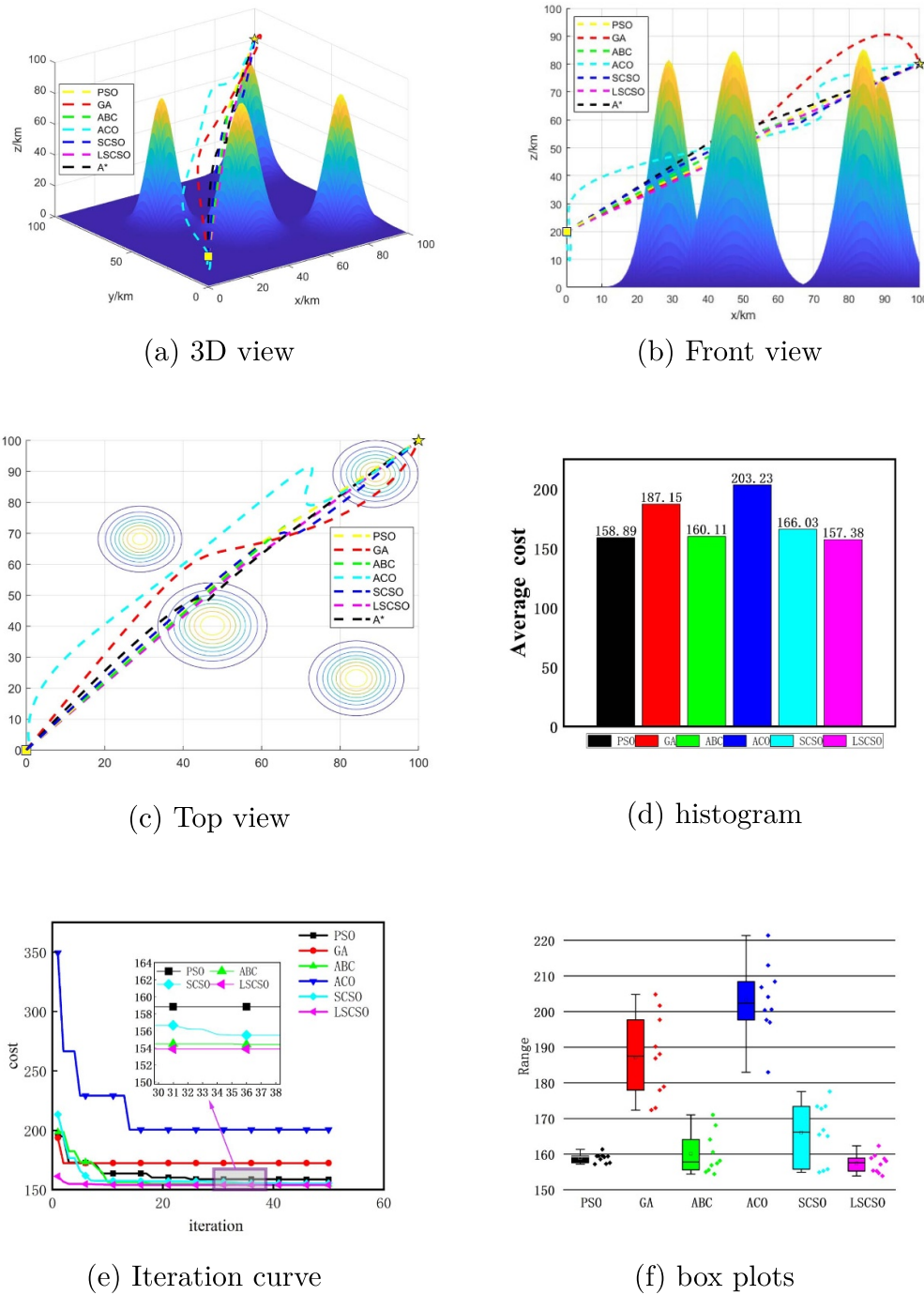


Figure 3. Experiment 1 data plot.

smooth flight path with minimal cost penalty. SCSO performs worse than LSCSO, with its inability to avoid unnecessary curvature, which leads to slow convergence, as can be seen in figure 3(d), its slow convergence prevents it from obtaining an optimal solution. A*, ACO, and GA exhibit similar performance, although both algorithms can plan a navigable path. However, the path cost of both algorithms is too high. It is clear from sub-figure (f) that the stability of LSCSO is also better than that of SCSO. The stability of PSO and ABC is commendable, but the cost results are poor.

5.2.2. High-dimensional experiments. In experiment 2 we chose to increase the terrain complexity by setting the starting point at (1, 1, 1) and the end point at (100, 100, 80) with a theoretical optimal value of 160.76. The number of peaks was raised to 30, with partial intersections between multiple peaks. To test the performance of LSCSO in complex terrain.

Table 3 gives the best value, mean and standard deviation of the six algorithms run independently for 10 times. Table 3 shows that LSCSO performance is the best. Although we learned from the data in Experiment 1 that LSCSO did not

Table 3. Experimental data.

Algorithm	Statistical data			Wilcoxon test		Time
	Avg	Opt	Std	P-value	Sign	
PSO	166.86	161.26	3.38	0.0455	+	11.77
GA	189.81	180.40	5.30	0.0001	+	11.29
A*	—	212.48	—	—	—	28.89
ABC	167.00	161.27	5.77	0.0412	+	11.26
ACO	203.81	197.51	6.48	0.0001	+	47.66
SCSO	180.64	161.37	9.96	0.0112	+	11.56
LSCSO	162.78	160.80	2.59	—	—	10.53

obtain the best standard deviation compared to PSO, when we use more complex terrain, we can clearly see that LSCSO is far superior to PSO in all performances. In other words, when the complexity of the terrain area and the dimensionality of the space increase, LSCSO can also plan shorter paths than all other algorithms, and the algorithm performance is more stable. Of course, more experiments will be added subsequently to verify the superiority of LSCSO in order to avoid experimental chance. Table 8 gives the error rates for the high-dimensional experiments, from which it can be seen that LSCSO is more advantageous in complex terrain with smaller error rates.

Figure 4 gives the path diagrams for each algorithm, similar to experiment 1. The performance of all the algorithms remains unchanged. ACO and GA still choose a detour, and their detours are much farther in the harsh environment. The A* algorithm does not perform well in complex terrain and even produces many bends. All the other algorithms plan relatively good paths, but only LSCSO plans a smooth curve that is close to a straight line, greatly reducing the cost of the flight, and it is also clear from subfigures (d) and (e) that LSCSO is optimal.

5.2.3. Randomized experiment. In this section, the above seven algorithms will be tested again in a 3D environment as a way to verify the superiority and stability of the improved algorithms. This experiment will use the method of randomly generating peaks with the following generation rules:

- (1) The starting point is (0, 0, 20), and the endpoint is (100, 100, 80).
- (2) The number of peaks is randomized in the range of (0, 50).
- (3) The center, height, and range of peaks are random. Range in (0, 100).

Table 4 gives the mean, optimum, and standard deviation of 10 independent runs of each algorithm in different environments. Where ‘*’ indicates that a safe and feasible path was not planned in all the 10 experiments. As can be seen from the table, A*, GA, and ACO do not perform well in all the experiments, PSO, ABC, and SCSO perform relatively well in low-dimensional environments, but are slightly weak in complex

environments. LSCSO, on the other hand, can plan the shortest path regardless of the environment and maintains stability and robustness in all six randomized experiments. Especially for Case 6, where most algorithms perform poorly, and some even fail to plan a safe path, LSCSO comes out on top in all data. This demonstrates the superiority of the improved algorithm.

In summary, LSCSO significantly outperforms its competitors in terms of convergence speed and solution quality under the same conditions. It shows even greater advantages in high-dimensional problems. These results are attributed to the proposed hybrid algorithm, i.e. the sand cat algorithm that incorporates various learning capabilities.

5.3. Comparative ablation experiments

In order to better test the effectiveness of the improvement factors, this paper conducts ablation experiments by performing the LSCSO algorithm. Three sub-versions of the LSCSO algorithm, SCSO-K, SCSO-L, and SCSO-P are proposed to test the effectiveness of the improvement method and its impact on the algorithm.

SCSO-K: Adds only the acoustic refraction strategy proposed in section 4.1 to the SCSO.

SCSO-L: Add the sonic refraction learning from section 4.1 and the learned behavioral strategies from section 4.2 to the SCSO.

SCSO-P: Add the sonic refraction learning from section 4.1 and the elite pool policy from section 4.3 to the SCSO.

To better compare the effect of each improved strategy, define the optimization rate:

$$\eta_{\text{avg}} = \frac{|l_m - l_0|}{l_0} = \frac{\Delta l}{l_0} \quad (25)$$

where l_m denotes the average fitness of the improved algorithm and average running time. l_0 denotes the average fitness of the unimproved algorithm and average running time.

5.3.1. Low-dimensional experiments. In this experiment, the starting path is set as (0, 0, 20) and the ending path is (100, 100, 80). Table 5 lists the experimental results for each algorithm, from which it can be seen that several sub-versions of the algorithm have been improved compared to the original algorithm. The cost optimization rate of SCSO-K is 3.6% and the average CPU time optimization rate reaches 9.4%. The optimization rate of LSCSO combining the three strategies reaches 5.2% and 8.2%, respectively. Not only that, the stability of each sub-version is improved over the original algorithm, which proves the superiority of the improved strategy.

This property is also corroborated by the experimental data plots given in figure 5. From subplot (f), it can be seen that SCSO has the largest box, its data has more discrete values, and is less stable, and its median is also above the other algorithms, which indicates that most of the adaptations of

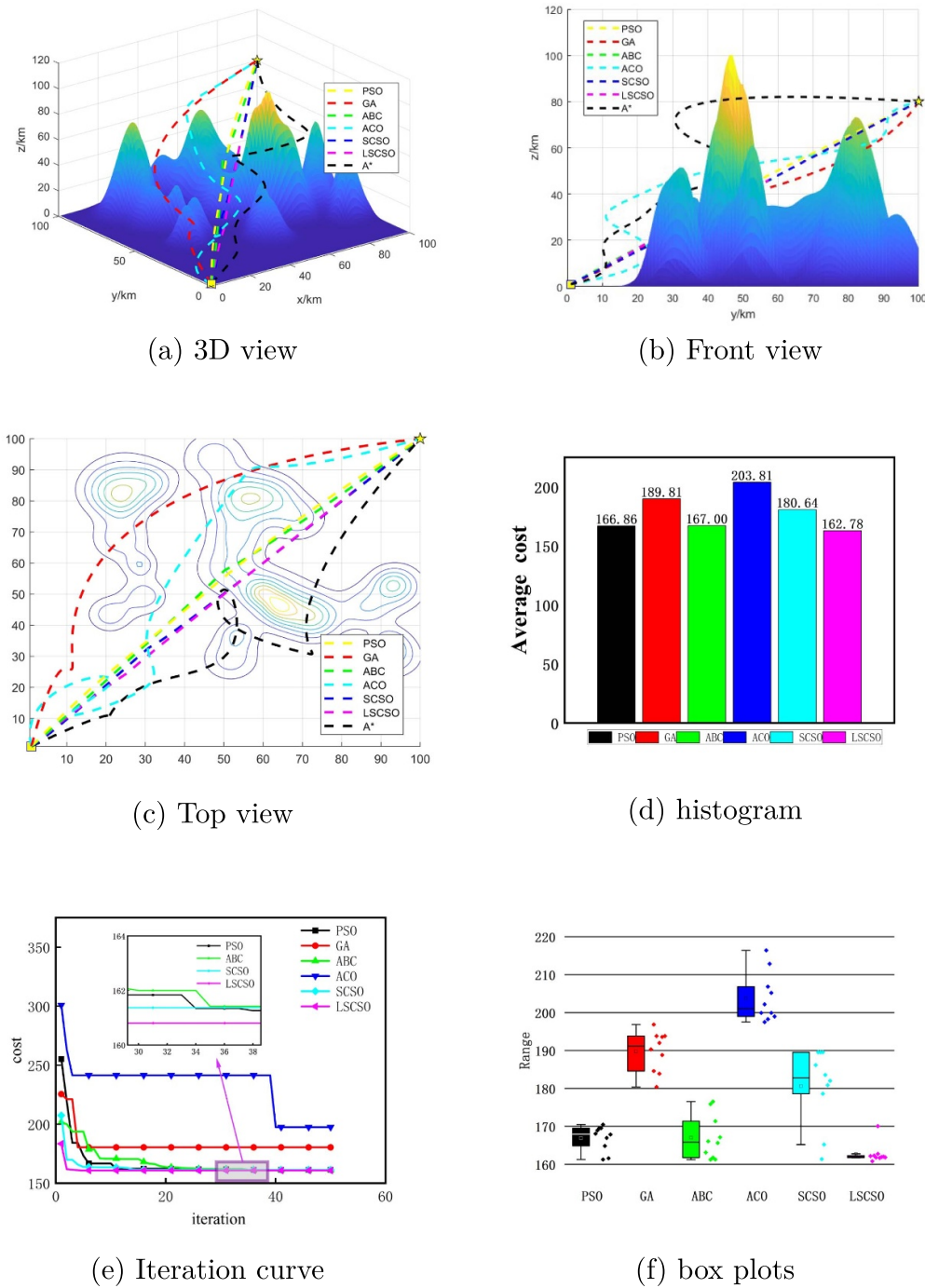


Figure 4. Experiment 2 data plot.

its algorithms are higher. From the iterative curve of sub-figure (d), it can be seen that the convergence speed of SCSO-P is very fast, which directly leads to the convergence speed of LSCSO as well, thanks to the core idea of SCSO-P is to accelerate the convergence speed by selecting the elite candidates and eliminating the poorly adapted populations. At the same time, learning also helps the population to obtain a better search space and avoid falling into a local optimum.

5.3.2. High-dimensional experiments. For the high-dimensional experiments, the starting point is set as (1, 1, 1) and the endpoint as (100, 100, 80), and each experiment is run independently ten times, and the experimental data of each algorithm is listed in table 6, and as in the low-dimensional experiments, each subversion still shows a good optimization seeking ability. It is worth mentioning that in the complex environment, the improvement of each algorithm is more obvious compared to the low-dimensional environment, and

Table 4. Experimental data.

Case	Measure	Algorithm						
		PSO	GA	A*	ABC	ACO	SCSO	LSCSO
Case1: 6peaks	Avg	158.04	176.91	—	157.79	207.52	163.82	154.62
	Opt	153.94	162.23	164.37	153.89	190.23	154.84	153.91
	Std	3.08	8.07	—	4.26	11.25	10.04	0.58
Case2: 8peaks	Avg	158.75	181.46	—	159.26	207.17	168.50	155.17
	Opt	155.49	162.97	171.73	154.12	186.87	154.80	153.97
	Std	4.38	11.65	—	4.17	12.23	16.00	1.55
Case3: 15peaks	Avg	163.15	186.44	—	166.42	217.41	169.28	155.61
	Opt	156.84	164.03	189.27	154.71	192.98	156.49	154.11
	Std	3.43	10.88	—	10.09	18.73	11.97	1.12
Case4: 24peaks	Avg	165.31	184.23	—	167.14	219.48	167.33	161.67
	Opt	160.63	164.98	202.93	161.96	202.08	156.91	155.69
	Std	5.06	11.81	—	4.80	10.40	5.69	5.04
Case5: 33peaks	Avg	170.21	219.32	—	172.52	230.12	192.24	164.52
	Opt	159.14	188.43	227.39	158.80	216.49	162.79	156.10
	Std	8.72	28.99	—	16.00	13.98	25.17	8.58
Case6: 43peaks	Avg	197.79	*	—	182.37	*	214.27	173.00
	Opt	184.64	*	264.57	166.24	*	170.03	160.25
	Std	10.20	*	—	10.94	*	32.69	7.86

Table 5. Experimental data.

Algorithm	Statistical data			Wilcoxon test			Rate	
	Avg	Opt	Std	P-value	Sign	Time	Cost	Time
SCSO-K	160.01	153.99	4.01	0.0489	+	9.00	3.6%	9.4%
SCSO-L	163.48	155.25	5.71	0.0420	+	9.14	1.5%	7.2%
SCSO-P	164.07	153.92	6.44	0.0445	+	9.12	1.2%	7.4%
SCSO	166.03	155.28	8.39	—	—	9.85	—	—
LSCSO	157.38	153.87	2.62	0.0312	+	9.04	5.2%	8.2%

the optimization rate of each optimization is generally higher than 5%, which shows that the improved algorithms have this stronger performance in the complex and multi-dimensional environment, and are suitable for complex environments.

Figure 6 also presents the experimental data plots of each algorithm, and from subplot (f), it can be seen that SCSO-L and SCSO-P are comparable to SCSO in terms of stability, but their cost is lower than SCSO. SCSO-K outperforms SCSO in terms of stability, while LSCSO is the best in terms of both adaptability and stability, which indicates that LSCSO inherits not only the optimization-seeking ability of SCSO-L and SCSO-P but also the stability of SCSO-K.

5.3.3. Randomized experiment. The experiments in this section continue the generation rules of the randomized experiments in experiments in section 5.2.3 and test each sub-version, and table 7 gives the average cost, optimal value, standard deviation, and cost optimization rate of each algorithm. From the table, it can be seen that each sub-version has different degrees of improvement in different environments, and the LSCSO that combines the three strategies has the largest improvement, and this improvement is especially obvious in high-dimensional environments. This shows the effectiveness and superiority of the improved algorithm.

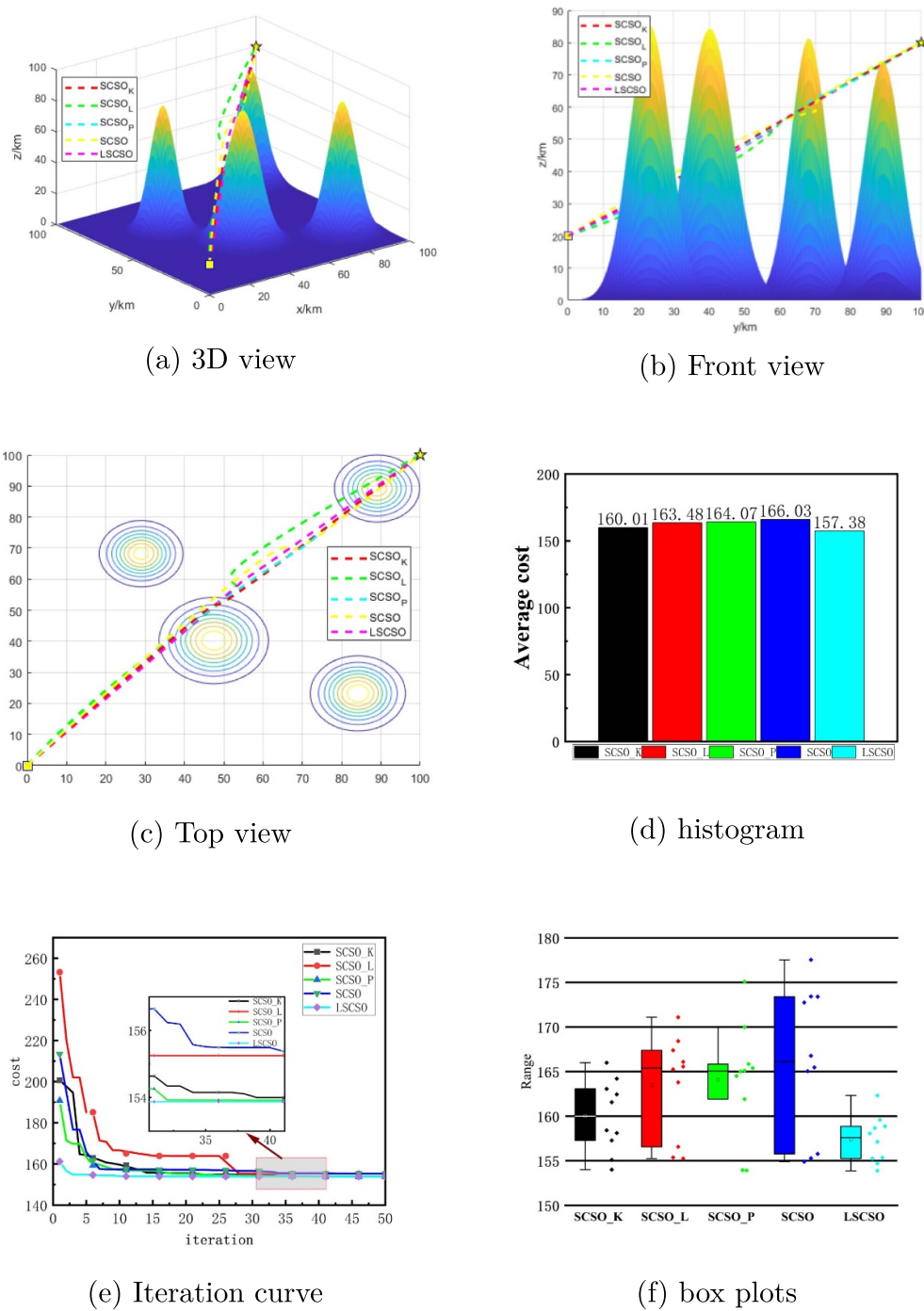
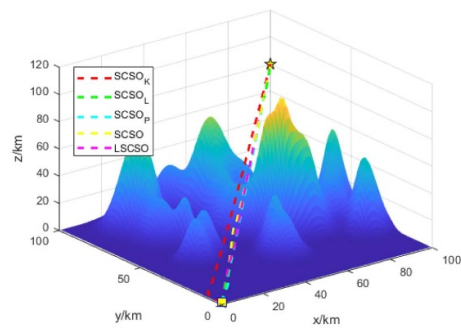


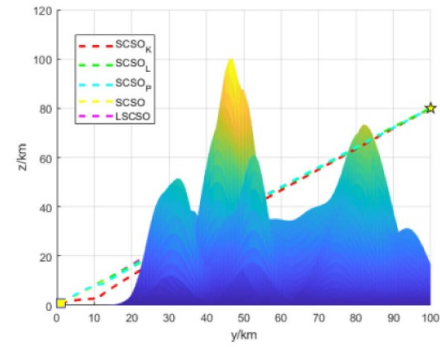
Figure 5. Experiment 3 data plot.

Table 6. Experimental data.

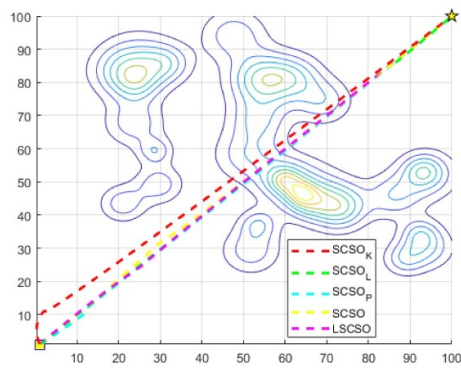
Algorithm	Statistical data			Wilcoxon test		Rate		
	Avg	Opt	Std	<i>P</i> -value	Sign	Time	Cost	Time
SCSO-K	170.34	164.15	5.17	0.0171	+	10.83	5.7%	6.3%
SCSO-L	169.59	162.53	6.92	0.0139	+	10.58	6.1%	8.4%
SCSO-P	168.84	161.07	7.60	0.0057	+	10.53	6.5%	8.9%
SCSO	180.64	161.37	9.96	—	—	11.56	—	—
LSCSO	166.86	160.80	3.38	0.0112	+	10.46	7.6%	9.5%



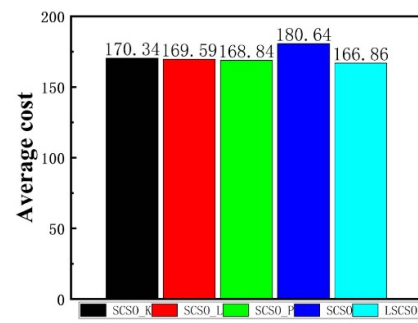
(a) 3D view



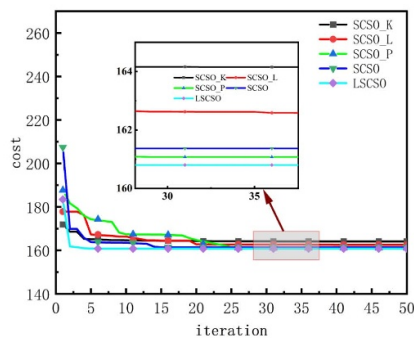
(b) Front view



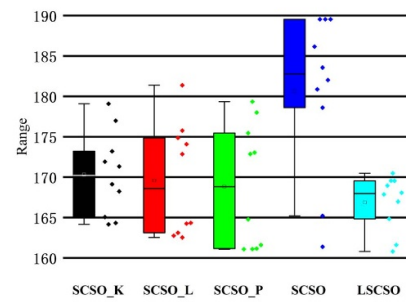
(c) Top view



(d) histogram



(e) Iteration curve



(f) box plots

Figure 6. Experiment 4 data plot.

Table 7. Experimental data.

Case	Measure	Algorithm				
		SCSO-K	SCSO-L	SCSO-P	SCSO	LSCSO
Case1: 3peaks	Avg	164.86	160.61	164.98	165.92	154.76
	Opt	153.94	154.89	154.28	154.84	153.79
	Std	8.91	6.72	7.40	11.85	0.72
	Rate	0.6%	3.2%	0.5%	—	6.7%
Case2: 7peaks	Avg	160.92	161.78	162.73	167.91	154.88
	Opt	154.18	154.78	155.08	155.65	153.93
	Std	7.32	8.84	6.33	14.58	0.58
	Rate	4.1%	3.7%	3.1%	—	7.8%
Case3: 14peaks	Avg	165.61	165.60	164.87	169.60	159.52
	Opt	155.98	156.58	156.49	159.63	154.58
	Std	7.87	5.33	6.41	11.63	3.65
	Rate	2.4%	2.4%	2.8%	—	5.9%
Case4: 22peaks	Avg	172.98	170.50	173.05	175.89	160.72
	Opt	156.69	158.68	161.79	165.80	154.76
	Std	8.34	5.92	5.18	12.01	4.28
	Rate	1.7%	3.1%	1.6%	—	8.6%
Case5: 31peaks	Avg	186.56	173.69	178.11	195.05	163.76
	Opt	169.39	167.24	166.55	170.25	157.17
	Std	13.96	5.52	9.90	24.01	5.43
	Rate	4.4%	10.9%	8.7%	—	16.0%
Case6: 40peaks	Avg	191.47	188.76	189.63	213.25	179.46
	Opt	170.32	171.56	168.67	178.44	158.30
	Std	15.18	13.36	12.84	33.03	7.98
	Rate	10.2%	11.5%	11.1%	—	15.8%

Table 8. Experimental error rate.

Algorithm	Low-dimensional	High-dimensional
	Error rate	Error rate
PSO	3.1%	0.3%
GA	12.2%	12.2%
A*	3.9%	32.1%
ABC	0.5%	0.3%
ACO	28.7%	22.8%
SCSO	1.1%	0.4%
LSCSO	0.16%	0.02%

6. Conclusion and future works

UAV path planning is an important part of the UAV mission system, in order to solve the UAV path planning problem, this paper proposes a sand cat algorithm incorporating learning behaviors. LSCSO incorporates refraction learning and elite pooling strategies into SCSO, effectively increasing the convergence speed of the algorithm as well as expanding the population search range. In addition, a new location-updating approach inspired by the behavior of sand cats is proposed, which significantly increases the convergence accuracy of the algorithm and its ability to jump out of the local optimum. Furthermore, a 3-times B-spline curve was used to generate smooth paths. Comprehensive experiments were conducted

to test the performance of the proposed algorithms. The results show that LSCSO is greatly superior to the compared algorithms in all the cases. These show the positive impact of several improvement strategies on LSCSO. It can be used as a safe and effective solution for UAV path planning.

However, this study may also have some limitations. For example, the improvement methods in this paper are tailored to address the shortcomings of SCSO. It may be possible that combining it with other heuristic algorithms will not improve the performance. In addition, the real flight environment in UAVs is very dangerous and complex. The terrain model presented in this paper is not a universally applicable model and the performance of LSCSO needs to be tested with a larger sample of models. In a follow-up study, we will increase the sample size to test the performance of LSCSO under different environmental constraints. Due to the limitations of current technology and theory, we cannot fully address the above limitations, but we still have some outlooks for future research: The meta-heuristic algorithm has the advantages of fast operation and high efficiency, but it is difficult to guarantee the performance of the original solution method when the flight model and constraints change. Reinforcement learning is a current research hotspot, and it has the advantage of being able to interact with different environments to make optimal choices, but it does take a lot of time to learn. There is little literature combining deep learning with heuristic algorithms to solve the UAV path planning problem. In future research, as deep learning techniques become more mature, the effective combination of the two may provide more universally applicable solutions for UAV path planning in different environments.

Data availability statement

The data cannot be made publicly available upon publication because no suitable repository exists for hosting data in this field of study. The data that support the findings of this study are available upon reasonable request from the authors.

Funding

This work was supported by the National Natural Science Foundation of China (Grant Nos. 21466008, 21566007, 21968008) and the Guangxi Natural Science Foundation (Grant No. 2019GXNSFAA185017).

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